

# Nominal Budgets and Realized Resources in Closed-API Large-Language-Model Inference: A Performance Evaluation Protocol

Soroush Vahidi<sup>a,1,\*</sup>

<sup>a</sup>*Ying Wu College of Computing, New Jersey Institute of Technology, Newark, New  
Jersey, USA*

---

## Abstract

Nominal inference budgets are convenient for large language model (LLM) systems, but equal logical entitlement need not imply equal successful completions, retries, tokens, latency, cost, or selector attribution. We present a protocol for closed-API, budgeted inference that separates discovery from commit-rule selection, evaluates identical-pool controls, records a componentwise realized resource vector, labels provider-transfer evidence, and preserves protocol-blocked outcomes. On a 15-cell provider-by-dataset matrix ( $n=3394$ ; Fireworks $\times$ GPQA-Diamond protocol-blocked), identical-pool plurality over four generators (Pooled-4) exceeds a failure-trace gated selector (FTA): 66.53% versus 65.00% (McNemar  $p=0.00027$ ). The dominant FTA gate reverses sign under provider transfer, including 2/18 and 7/31 rescue/regression patterns on Azure mathematics cells. A resource audit changes efficiency interpretation without changing accuracy: reconstructed Frontier successful completions rise from 2.78 to 5.38 of nominal budget  $B=6$ , while token/USD fields remain lower bounds. The contribution is the protocol: nominal budgets alone are insufficient for defensible closed-API inference-performance conclusions.

*Keywords:* budgeted inference, performance measurement, resource accounting, closed-API inference, large language models

---

\*Corresponding author

*Email address:* [sv96@njit.edu](mailto:sv96@njit.edu) (Soroush Vahidi)

<sup>1</sup>Sole author. ORCID: 0000-0003-1934-6282.

---

## 1. Introduction

Budgeted test-time inference is often summarized by one performance number: final accuracy under a shared nominal budget of samples, branches, or reasoning actions. That number becomes ambiguous when the nominal budget is treated as the resource measure itself and when discovery, selection, and accounting are collapsed into a single outcome:

- Did any correct answer enter the candidate pool?
- Did the commit rule select a correct answer when one was present?
- Did compared methods share an identical pool, or did one simply receive more diverse candidates?
- Does a failure-derived selection rule that helps under one commercial API transfer to another?
- Does a matched nominal budget  $B$  imply comparable calls, tokens, latency, or cost?

These questions matter especially for closed-API, multi-generator evaluation: providers differ, telemetry fields may be incomplete, and post-generation rules can appear effective until compared to plurality on the *same* extracted answers. Prior work already separates coverage from selection and studies consensus aggregators [1, 2, 3, 4], routers, and resource reporting [5]. The measurement gap is their *joint* operationalization under one closed-API, multi-provider, budgeted protocol with paired identical-pool controls, transfer analysis, provenance labels, and instrumentation audit.

*Contribution.* This article contributes a controlled performance-evaluation protocol for budgeted closed-API inference (Figure 1). The protocol combines identical-pool attribution, provider-transfer analysis, protocol-blocked outcomes, and componentwise realized-resource accounting under incomplete closed-API telemetry. The empirical sections use that protocol to show four consequences: identical-pool plurality is a required selector control, failure-mined rules need transfer labels, nominal budgets are not realized resource vectors, and provenance/blocked cells affect interpretation even when accuracy values are unchanged.

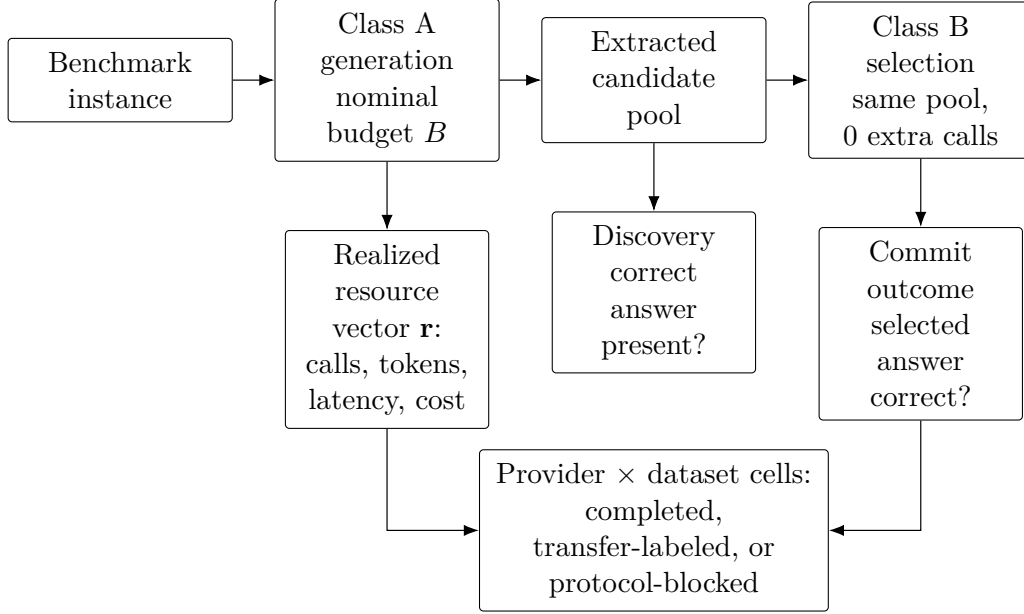


Figure 1: Protocol schematic. Nominal budget  $B$  defines the logical action contract; realized resources are measured separately. Selection claims are attributed after candidate availability is fixed, and provider $\times$ dataset cells may be completed, transfer-labeled, or protocol-blocked.

*Roadmap.* Section 2 positions the work relative to performance measurement, test-time compute, routing, consensus, and verifier literature. Sections 3–5 define the evaluation objects and closed-API workload. Sections 6–9 report the matrix, identical-pool selection tests, provider transfer, and resource accounting. Section 10 reports offline counterfactuals, and Sections 11–13 summarize scope and implications for performance-evaluation practice.

*Measurement objects.* The named procedures are measurement objects for exercising the protocol (Section 4). Class A single-generator policies under  $B=6$  include a reserved-attempt full-expansion diagnostic (Frontier) and closed-API adapters inspired by L1, S1, and TALE [6, 7, 8]. Class B post-generation selectors over the shared pool include Pooled-4 and FTA. Offline oracles are upper bounds only. A single-cell Azure $\times$ GSM8K call/sample-budget control compares Frontier to same-model self-consistency with  $N=6$  (SC- $N=6$ ; Section 6.3).

## 2. Related Work

*Performance-measurement and benchmarking methodology.* Performance-evaluation practice emphasizes construct validity, measurement semantics, and protocol transparency: reported rankings depend on what is counted, at what layer, and under what workload definition [9, 10, 11]. In budgeted LLM inference, the analogous construct-validity issue is whether a nominal action budget measures the resource whose performance claim is being made. We therefore separate the nominal budget from successful completions, HTTP retries, generated tokens, observed latency, and estimated dollars. Recent surveys of test-time scaling likewise recommend reporting resource dimensions beyond final accuracy under nominal budgets [12, 5]. Our protocol follows that measurement view by reporting componentwise realized resources and by auditing instrumentation artifacts that can alter efficiency interpretation without altering correctness outcomes.

Recent systems-performance work reinforces the same scope for cloud and AI services: cloud benchmarking methodology calls for explicit measurement and reporting practice; PEVA work on cloud microservices studies uncertain latency–resource relationships under dynamic workloads; and PEVA work on serverless ML treats public-cloud scaling and cost as part of the performance object [13, 14, 15]. We adapt that systems-performance view to closed-API LLM inference, where providers expose incomplete telemetry and nominal budgets are not realized resource vectors.

*Benchmark validity and serving-resource context.* LLM evaluation infrastructure such as HELM emphasizes scenario coverage, multi-metric reporting, and released evaluation artifacts, while LiveBench addresses benchmark drift and contamination through frequently updated, objectively scored workloads [16, 17]. GSM-Symbolic further shows that GSM8K-style math scores can be sensitive to symbolic perturbations of the same underlying problem templates [18]. These works motivate our workload-validity caveats rather than serving as direct experimental baselines. In parallel, open serving-stack work such as vLLM/PagedAttention optimizes KV-cache memory and throughput under instrumented serving systems [19]; our setting instead measures closed APIs whose telemetry is partial and whose nominal call budgets do not expose a complete resource vector.

*Workload characterization and test-time controls.* Workload characterization asks whether compared systems face the same demand process and observable

resource contract. Adaptive compute-allocation work shows that how budget is spent can matter as much as model size [5, 20, 21, 22]. Length-controlled and budget-forcing approaches intervene directly on inference trajectories [6, 7, 8]. We treat these mechanisms as workload controls in a shared nominal- $B$  environment and evaluate them under common extraction and scoring conventions (Table 1).

*Discovery, consensus, and selection attribution.* Pass@ $k$  analyses separate whether any sample is correct from whether a system selects it [1, 2]. Majority vote is the standard same-model self-consistency (SC) aggregator [4]; universal SC extends aggregation beyond extractable answers [23]. Adaptive and early-stopping SC variants exist [24, 25, 26, 27]. Best-of- $N$  and process verifiers supply alternative selection signals [28, 29]. Fixed-pool oracle-gap diagnostics decompose recoverable mass and harm [3]. Cascades and routers choose among models [30, 31, 32]. Relative to this literature, we keep the generation–selection distinction as inherited, and instead ask how identical-pool plurality, failure-mined deferral, and componentwise realized resources interact under one closed-API multi-generator protocol. Pooled-4 targets a *heterogeneous* four-generator pool—a different axis from same-model SC—with one Azure×GSM8K SC- $N=6$  budget cell as a supporting control (Section 6.3).

*Resource accounting in distributed/closed inference services.* Closed inference services expose partial telemetry: successful calls, retries, token fields, latency, and billable cost may not align perfectly. This complicates cross-policy efficiency comparisons even under matched nominal call budgets, and it creates a distributed-service benchmarking problem rather than only an LLM-algorithm problem. Our instrumentation treatment aligns with the performance-measurement principle that metrics must be interpreted at the layer where they are observed, with reconstruction status disclosed explicitly.

*Tree search and stopping.* Tree-of-Thoughts expands intermediate states under search heuristics [33]; classical metareasoning frames stopping as value-of-computation [34]. The evaluated Frontier configuration does not test adaptive early stopping (Section 4.2). Historical explorations outside the evaluated protocol are documented only in the supplementary material.

### 3. Evaluation Framework

*Setting.* The protocol deliverable is a set of accounting and attribution rules for budgeted closed-API inference. An instance  $x_i$  has gold answer  $g_i$ , used offline only for evaluation. A single-generator method may issue at most  $B$  logical reasoning actions (provider calls) while producing a committed answer. We call  $B$  the *nominal budget*: the logical entitlement specified by the experiment, not an observed resource total. A candidate pool  $C_i = \{y_{i,1}, \dots, y_{i,m}\}$  is a multiset of extracted final answers under a shared protocol. A selector  $f$  maps  $C_i$  and optional gold-free metadata to a prediction  $\hat{y}_i$ .

*Discovery and selection.* Following coverage / pass@ $k$  analyses [1, 2] and fixed-pool recoverable-mass diagnostics [3], define the discovery indicator

$$D_i = \mathbf{1}\{\exists y \in C_i : y \equiv g_i\}, \quad (1)$$

i.e., whether a correct candidate is present in the evaluated pool. With  $n$  scored instances and  $N_D = \sum_i D_i$ , define discovery rate (DR), unrecoverable failure rate (UFR), conditional selection accuracy ( $\text{CSA}_f$ ), and selection failure rate ( $\text{SFR}_f$ ):

$$\begin{aligned} \text{DR} &= \frac{1}{n} \sum_i D_i, & \text{UFR} &= 1 - \text{DR}, \\ \text{CSA}_f &= \frac{\sum_i \mathbf{1}\{\hat{y}_i \equiv g_i\} D_i}{N_D} \quad (N_D > 0), & \text{SFR}_f &= 1 - \text{CSA}_f. \end{aligned} \quad (2)$$

Blocked or unscorable instances are excluded from  $n$  rather than counted as  $D_i = 0$  (Section 5). For pool-only selectors under exact-match scoring,

$$\text{Acc}_f = \text{DR} \times \text{CSA}_f, \quad (3)$$

because such a selector can be correct only when  $D_i = 1$ .

*Identical-pool consensus.* Let  $\text{Cons}(C_i)$  denote plurality over  $C_i$ , with ties broken by a fixed documented rule. An *identical-pool control* holds  $C_i$  fixed and changes only the commit rule. Consensus regret  $\text{CR}(f) = \text{Acc}(\text{Cons}) - \text{Acc}(f)$  is positive when consensus beats  $f$  on the same pools. We report CR whenever a nontrivial selector is compared [4, 23]. Offline oracles over  $C_i$  are Class C upper bounds only [3].

*Realized compute vector.* Nominal budget  $B$  is an experimental contract, not a realized-resource metric. It counts logical Class A reasoning actions, not dollars or FLOPs. A *successful completion* is a provider call that returns a usable model output on the reported path; a *retry* is an internal repeated HTTP attempt after a transient or malformed response. Under the nominal budget we record a componentwise realized resource vector

$$\mathbf{r} = (\text{calls}, \text{generated tokens}, \text{latency}, \$), \quad (4)$$

distinguishing successful completions from retries, generated-token fields, observed latency, and estimated dollar cost. We do not collapse  $\mathbf{r}$  to one scalar. For post-generation selectors, pool cost is the joint cost of the generators that form  $C$ , plus zero marginal selector calls. Equal  $B$  equalizes Class A opportunity, not tokens, latency, or dollars (Section 9).

### 3.1. Evaluation objects

Equations (1)–(4) reuse established constructs. The evaluation objects of this study are the joint reporting of DR, CSA, CR, and componentwise  $\mathbf{r}$ ; the separation of Class A (allocation under  $B$ ) from Class B (selection over a shared pool); development-environment labels for failure-mined rules; and blocked cells as *protocol-blocked outcomes*, meaning bounded-protocol failures reported separately from scored accuracy. Section 4 specifies the procedures used to instantiate those objects.

## 4. Measurement Instruments

The procedures below instantiate the evaluation questions of Sections 1 and 3: how nominal budgets map to realized resources; how much of a result is discovery versus commit-rule selection; and whether failure-mined commit semantics transfer across providers. The hierarchy is evaluation question  $\rightarrow$  control/instrument  $\rightarrow$  evidence. Names such as Frontier, FTA, Pooled-4, and the adapters identify those controls in tables.

### 4.1. Taxonomy

- **Class A — single-generator methods (nominal  $B=6$  each):** Frontier; the L1-style, S1-inspired, and TALE-EP-style closed-API adapters.

- **Class B — selectors over the shared multi-generator pool:** Pooled-4 (plurality); FTA (failure-trace gated deferral). Zero *additional* model calls after pool generation; pool construction may cost up to  $4B$  nominal generator actions.
- **Class C — offline oracles:** coverage upper bounds. Never treated as peer winners.

Nominal budget  $B$  is the logical entitlement of Section 3; realized  $\mathbf{r}$  is tracked separately.

#### 4.2. Frontier (Class A diagnostic)

Frontier is the Class A accounting diagnostic: a reserved-attempt full-expansion policy with fixed parameters across cells. It runs two direct-chain attempts, forms a gold-free incumbent by plurality, then uses the remaining nominal budget for challenger expansion/verification before applying a fixed gold-free override rule. On preserved validated records, reconstructed successful calls average 5.38 of nominal 6 (Section 9).

The implementation template retained an early-commit gate with sentinel thresholds  $(2.0, 2.0, -1.0)$ . Because top support is a vote-share in  $[0, 1]$ , the gate never fires; reported runs therefore do not evaluate adaptive early stopping. Full pseudocode and action semantics appear in Appendix F.

#### 4.3. Class B selectors

*Pooled-4.* Plurality over {Frontier, L1, S1, TALE} (agreement  $\geq 2$ ; ties fall back to Frontier). Same extracted answers as FTA; no model calls.

*FTA (gated deferral diagnostic).* FTA retains Frontier’s answer or defers using Frontier trace metadata. Gates fire in precedence order; at most one per instance (Appendix G):

- **Low-Depth Guard (FIX-2).** On a single-weak-branch (or support-zero) signature, replace Frontier by majority among {L1, S1, TALE}.
- **External-Consensus Fallback (FIX-4).** Else, if Frontier committed via direct incumbent agreement and all three externals unanimously disagree, adopt that unanimous answer.

FTA is the transfer diagnostic for failure-mined commit semantics. Gate rules were developed on Cohere×GSM8K failure analysis; that cell is development-adjacent (Section 8). Runtime features are gold-free (Appendix A).



#### 4.4. Bounded answer repair

A deterministic support-based re-ranking is computed for every method but overridden by Frontier’s controller-committed answer when present, while L1/S1/TALE typically score the repair layer. Offline quantification and winner-stability checks appear in Appendix B; primary tables retain the historical scoring convention.

### 5. Experimental Protocol

#### 5.1. Providers, datasets, and shared protocol

Providers and exact hosted model IDs:

- Cohere: `command-r-plus-08-2024`.
- Azure OpenAI: `gpt-4.1-mini`.
- Google Vertex Gemini: `gemini-2.5-flash`.
- Fireworks: `accounts/fireworks/models/deepseek-v4-pro`.

Datasets and sizes per completed cell:

- GSM8K [35]:  $n=300$  examples under seed 71 and nominal  $B=6$ .
- MATH-500 [36, 29]: the 500-problem MATH subset popularized in process-supervision work; we evaluate  $n=300$  examples under the same seed/ $B$  protocol.
- GPQA-Diamond [37]:  $n=198$  (full Diamond set under our MCQ scoring protocol).
- StrategyQA [38]:  $n=100$ . This sample is lower-powered for detecting small effects; we treat small StrategyQA deltas as descriptive, not as strong evidence.

Total paired examples across completed cells: 3,394. The evaluated protocol uses a single nominal budget  $B=6$ .

**Answer scoring.** Final answers use dataset-appropriate normalization. Gold labels are offline-only. Closed commercial APIs; public-benchmark contamination cannot be ruled out, so absolute accuracies are protocol snapshots. Defaults unless overridden: temperature 0.1, max output tokens 220, up to 5 API retries (base 1s, backoff  $\times 2$ , max 20s, jitter 0.5s).

### 5.2. Matrix completeness and protocol-blocked outcome

Fifteen of sixteen provider×dataset cells completed. Fireworks×GPQA-Diamond is `BLOCKED_PROTOCOL_NONCONVERGENCE`: after parser/schema and token-floor repairs, the Fireworks-hosted model still exhausted its bounded generation budget on a material subset without producing a scorable MCQ answer. We record this as a protocol-level outcome: it is not treated as zero accuracy, not imputed, and not silently omitted (Section 6).

### 5.3. Adapter fidelity and provenance

L1/S1/TALE comparisons use closed-API adapters, not official training/serving reproductions (Table 1; Appendix C). Class A numerical attributions are to the named adapters. Azure×GPQA FTA is offline selector replay on stored generations. Trust tiers (Appendix D): Tier A denotes Cohere/Azure/Vertex×GSM8K cells with the strongest draw and scoring provenance; Tier B denotes the remaining completed cells. An earlier fallback-prompt SC attempt is excluded from all results; the Azure true same-model SC control records question-source hashes (Section 6.3).

Table 1: Baseline fidelity disclosure. None is an official training/serving reproduction.

| Evaluated label | Original paper            | Code                          | Classification                                     |
|-----------------|---------------------------|-------------------------------|----------------------------------------------------|
| L1 adapter      | Aggarwal & Welleck (2025) | <code>cmu-13/11</code>        | Prompt surface only; no LCPO training/model        |
| S1 adapter      | Muennighoff et al. (2025) | <code>simplescaling/s1</code> | Chat “Wait” continuation vs. token-stream forcing  |
| TALE-EP adapter | Han et al. (2024)         | <code>GeniusHTX/TALE</code>   | Character heuristic vs. zero-shot budget estimator |

*Repair asymmetry*: bounded repair is overridden by Frontier’s controller-committed answer but typically surfaces for L1/S1/TALE. Frontier’s scored answer differs from its repair field on 151/3394 rows; external winner flips under the repair counterfactual: 0 (Appendix B).

### 5.4. Reproducibility levels

We distinguish three reproducibility claims. *Public aggregate-record verification*: released aggregate audit records, public manifests, source tables, and analysis scripts support aggregate-record verification of the reported tables and numerical invariants. *Historical API-output regeneration*: proprietary, stochastic closed-API services need not reproduce identical completions under the same public model ID. *Historical subset reconstruction*: for some Tier B cells the evaluated subset is preserved in aggregate form, but the complete original draw procedure cannot be reconstructed from the released

Table 2: Provider×dataset landscape under nominal  $B=6$ . Class A columns are closed-API generators (Frontier diagnostic; L1-/S1-/TALE-style adapters). Class B columns select over the *same* four-generator pool (pool cost  $\leq 4B$ ; 0 extra model calls). Bold marks the highest Class A count and, separately, the higher of FTA vs. Pooled-4 as descriptive highlighting. Trust tiers and the Fireworks×GPQA blocked outcome are defined in Section 5.

| Provider  | Data       | Front.                          | L1             | S1            | TALE           | FTA            | Pooled-4       |
|-----------|------------|---------------------------------|----------------|---------------|----------------|----------------|----------------|
| Cohere    | GSM8K      | 230/300                         | <b>249/300</b> | 246/300       | 235/300        | 260/300        | <b>261/300</b> |
| Cohere    | MATH-500   | <b>87/300</b>                   | 73/300         | 84/300        | 76/300         | 92/300         | <b>98/300</b>  |
| Cohere    | GPQA-D     | 68/198                          | 74/198         | 69/198        | <b>77/198</b>  | <b>72/198</b>  | <b>72/198</b>  |
| Cohere    | StrategyQA | 78/100                          | 69/100         | 77/100        | <b>83/100</b>  | <b>76/100</b>  | <b>76/100</b>  |
| Azure     | GSM8K      | <b>276/300</b>                  | 269/300        | 246/300       | 204/300        | 260/300        | <b>276/300</b> |
| Azure     | MATH-500   | <b>167/300</b>                  | 142/300        | 161/300       | 112/300        | 143/300        | <b>164/300</b> |
| Azure     | GPQA-D     | 103/198                         | <b>105/198</b> | 104/198       | 94/198         | 102/198        | <b>104/198</b> |
| Azure     | StrategyQA | 70/100                          | 72/100         | <b>76/100</b> | 74/100         | <b>72/100</b>  | <b>72/100</b>  |
| Vertex    | GSM8K      | <b>279/300</b>                  | 262/300        | 244/300       | 255/300        | 272/300        | <b>278/300</b> |
| Vertex    | MATH-500   | 217/300                         | 220/300        | 210/300       | <b>223/300</b> | <b>228/300</b> | 224/300        |
| Vertex    | GPQA-D     | <b>113/198</b>                  | 106/198        | 108/198       | 110/198        | <b>125/198</b> | 123/198        |
| Vertex    | StrategyQA | <b>84/100</b>                   | 82/100         | 83/100        | <b>84/100</b>  | <b>84/100</b>  | <b>84/100</b>  |
| Fireworks | GSM8K      | <b>223/300</b>                  | 210/300        | 160/300       | 191/300        | 224/300        | <b>230/300</b> |
| Fireworks | MATH-500   | 100/300                         | 105/300        | 94/300        | <b>114/300</b> | <b>117/300</b> | <b>117/300</b> |
| Fireworks | GPQA-D     | BLOCKED_PROTOCOL_NONCONVERGENCE |                |               |                |                |                |
| Fireworks | StrategyQA | <b>81/100</b>                   | 80/100         | 80/100        | 80/100         | <b>79/100</b>  | <b>79/100</b>  |

artifact record; that incompleteness limits independent regeneration of the sampling step, not verification of the reported aggregate analyses (Appendix I). AI-assisted code inspection, code modification, and analysis organization were human-reviewed; all reported outputs were independently verified using frozen scripts, records, and manifests.

## 6. Matrix Landscape: Discovery, Adapters, and Scope

Table 2 is a workload landscape, not an efficiency or selector-attribution result. Under nominal budget  $B=6$ , it records Class A generators and Class B identical-pool selectors across providers and datasets; Section 5 defines the unscored Fireworks×GPQA protocol-blocked outcome. Sections 7–9 then isolate the comparisons the matrix alone cannot settle: identical-pool selection, provider transfer, and realized-resource accounting. Bounded repair follows the historical convention in Section 4.4; Appendix B verifies that a uniform-repair counterfactual causes no external winner flips.

### 6.1. Does higher discovery predict safer selection?

Absent-from-pool errors are unrecoverable by pool-only selectors; present-but-misselected errors are the only errors a commit rule can still fix (Equation (3)). Writing DR for discovery rate and CSA for conditional selection accuracy (Equation (2); Figure 1), higher discovery does not systematically track FTA gains: global Pearson  $r = -0.185$  (not significant at  $n=15$ ; unstable under leave-one-out; supplementary scatter diagnostic). Azure×GSM8K has near-ceiling discovery with a large negative FTA delta, while Vertex and Azure GPQA share essentially the same discovery rate with opposite selector signs (Section 8). Joint DR/CSA reporting is therefore required before identical-pool tests.

### 6.2. What do Class A adapter counts characterize?

Descriptively, within the evaluated closed-API adapter surfaces, the reserved-attempt full-expansion diagnostic is sole highest in 7/15 cells and tied highest in one more; remaining adapter surfaces occupy fewer cells. We use these counts to characterize the workload surface before the controlled selection and accounting analyses. An offline incumbent-only counterfactual loses to full Frontier in 9/15 cells (Section 10), indicating that challenger expansion is not inert under stored traces.

### 6.3. What does the single-cell same-model control establish?

Table 3 is a single-cell supporting control on Azure GPT-4.1-mini with GSM8K ( $n=300$ ): it matches call/sample budget between Frontier ( $B=6$ ) and true same-model fixed- $N$  self-consistency (SC; six samples). Each obtains 276/300 (92.00%; marginal exact binomial 95% CI 88.3304–94.8071%). Paired records show 298 ties and one method-only correct case each (two-sided exact McNemar  $p=1.0$ ; paired-bootstrap 95% percentile interval for Frontier–SC:  $[-1.00, +1.00]$  pp). No equivalence margin was prespecified. Panel B is contextual only.

## 7. Identical-Pool Selection

### 7.1. Does trace-conditioned selection beat identical-pool consensus?

Table 2’s Class B columns and Figure 2 test whether failure-trace gated deferral (FTA) adds value beyond plurality when candidate availability is held fixed. FTA and Pooled-4 operate on the same four-generator answers,

Table 3: Single-cell supporting control on Azure `gpt-4.1-mini`  $\times$  GSM8K ( $n=300$ ). Panel A matches call/sample budget (Frontier  $B=6$  vs. SC with six samples); rows are not token-, latency-, dollar-, or FLOP-matched. Panel B is contextual adapter/FTA context on the same IDs.

| Method                                                                        | Operational class            | Calls? | Pool/call basis                                             | Correct | Total | Accuracy |
|-------------------------------------------------------------------------------|------------------------------|--------|-------------------------------------------------------------|---------|-------|----------|
| <i>Panel A: direct paired same-model control (common 300 examples)</i>        |                              |        |                                                             |         |       |          |
| Frontier                                                                      | Structured generation        | Yes    | Nominal budget $B=6$                                        | 276     | 300   | 92.00%   |
| True same-model SC $N=6$                                                      | Fixed- $N$ generation + vote | Yes    | Exactly 6 samples                                           | 276     | 300   | 92.00%   |
| <i>Panel B: contextual adapter and replay rows (not a cost-matched class)</i> |                              |        |                                                             |         |       |          |
| L1-style adapter                                                              | Generation adapter           | Yes    | Nominal budget $B=6$                                        | 269     | 300   | 89.67%   |
| FTA replay selector                                                           | Offline pool selector        | No     | Shared four-generator pool ( $\leq 4B$ ); no selector calls | 260     | 300   | 86.67%   |
| S1-inspired adapter                                                           | Generation adapter           | Yes    | Nominal budget $B=6$                                        | 246     | 300   | 82.00%   |
| TALE-EP-style adapter                                                         | Generation adapter           | Yes    | Nominal budget $B=6$                                        | 204     | 300   | 68.00%   |

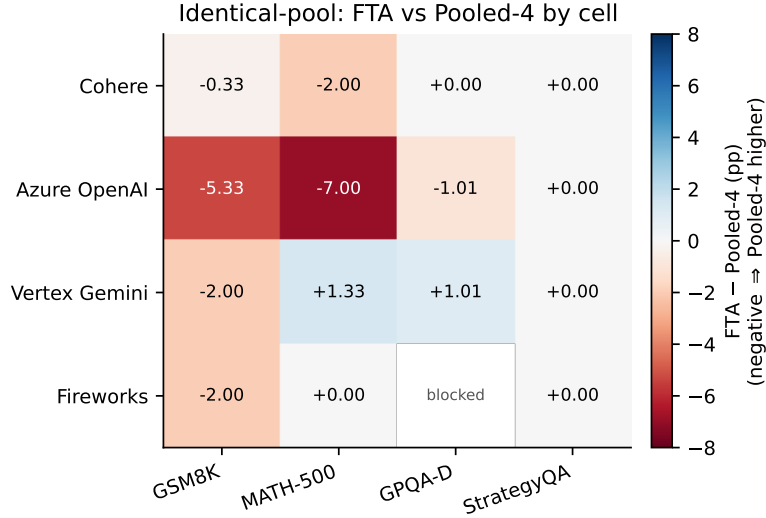


Figure 2: Per-cell difference in accuracy (FTA–Pooled-4, percentage points) on the *identical* four-generator pool. Negative values favor plurality. The comparison holds candidate availability fixed, so the map isolates commit-rule effects. Fireworks×GPQA is protocol-blocked.

so their paired difference isolates the commit rule rather than discovery. On that shared pool ( $n=3394$ ; 15 cells), Pooled-4 accuracy is 66.53% (2258/3394) and FTA is 65.00% (2206/3394):  $-1.53$  pp for FTA (McNemar  $p=0.00027$ ; discordant FTA/Pooled-4/tie counts 73/125/3196). A stratified within-cell paired bootstrap (10,000 replicates) places the Pooled-4–FTA gap at  $+1.53$  pp with 95% percentile interval  $[+0.74, +2.33]$  pp; the gap remains positive under leave-one-provider, leave-one-dataset, and leave-one-cell deletion. Omitting Azure reduces the gap to  $+0.52$  pp (McNemar  $p=0.30$ ), so the aggregate advantage is concentrated but directionally stable. Descriptively, Pooled-4 matches or exceeds FTA in 13/15 completed cells (most per-cell gaps are within resampling noise; bootstrap replicates reproduce  $\geq 13/15$  only  $\approx 17\%$  of the time). FTA is strictly higher only on Vertex×MATH-500 and Vertex×GPQA. A unique plurality exists on 88% of examples; the 12% Frontier-fallback subset is harder (26% vs. 74% accuracy). Alternative deterministic tie rules move overall accuracy by at most  $\sim 1.3$  points (Appendix K).

Positive consensus regret  $\text{CR}(\text{FTA})$  (Section 3) means this failure-derived

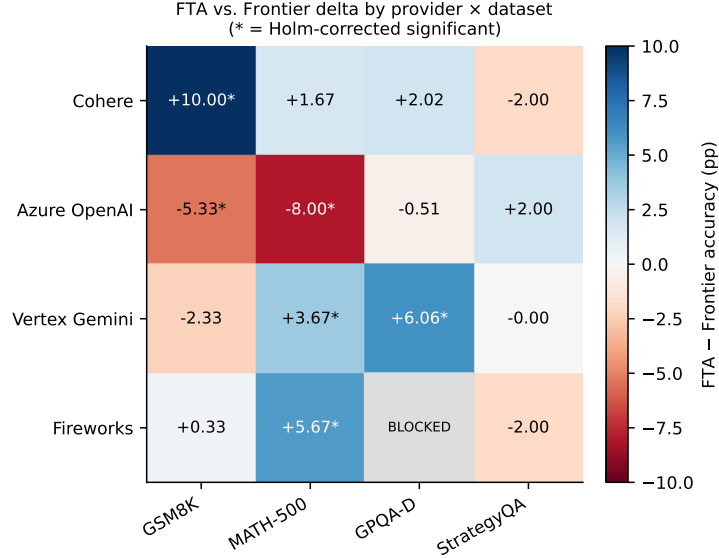


Figure 3: Per-cell FTA–Frontier accuracy delta. Asterisks mark Holm-corrected supported differences. Signs vary by provider; Azure has the only negative provider mean—transfer evidence for the commit rule, not a uniform selector ranking.

rule extracts less accuracy from the shared pool than plurality. Without that control, occasional FTA gains over a single Class A generator could be misread as selection progress. The measurement point is attribution: once candidate availability is fixed, this commit rule has positive consensus regret despite some generator-relative gains.

### 7.2. When FTA differs from its Class A generator, is the pattern transfer?

Relative to Frontier alone, FTA improves in 8/15 cells, regresses in 6/15, and ties in 1/15. After Holm-corrected paired McNemar tests, supported gains appear in four cells and supported regressions in two (Azure×GSM8K, Azure×MATH-500). Figure 3 treats that heterogeneity as transfer evidence for the commit rule.

*Diagnostic cells.* **Cohere×GSM8K** (Tier A; development-adjacent): high discovery (0.907); FTA improves over Frontier yet trails Pooled-4 slightly. **Azure×GSM8K**: discovery near ceiling (0.927) with FTA harming both Frontier and Pooled-4—selection failure under high discovery. **Vertex versus Azure GPQA**: matched discovery ( $\approx 0.636$ ) with opposite FTA signs.

Table 4: Leave-one-cell-out selector over 15 completed cells with coarse cell-level features. The selector chooses Pooled-4 in every fold and ties fixed Pooled-4; the oracle row is non-deployable headroom.

| Policy                              | Correct | Total | Accuracy | Deploy. | Outcome use |
|-------------------------------------|---------|-------|----------|---------|-------------|
| Held-out learned policy             | 2258    | 3394  | 66.53%   | Yes     | No          |
| Best training-selected fixed method | 2258    | 3394  | 66.53%   | Yes     | No          |
| Fixed Pooled-4                      | 2258    | 3394  | 66.53%   | Yes     | No          |
| Frontier                            | 2176    | 3394  | 64.11%   | Yes     | No          |
| FTA replay selector                 | 2206    | 3394  | 65.00%   | Yes     | No          |
| Oracle cell selector                | 2287    | 3394  | 67.38%   | No      | Yes         |

StrategyQA ( $n=100$ ) effects are near zero and are not treated as strong evidence.

### 7.3. Can cell-level selection improve over a fixed policy without leakage?

We ask whether a leakage-free selector calibrated only on training cells can choose among a frozen method set for a held-out provider–dataset cell. The design uses leave-one-cell-out over the 15 completed cells; candidate methods are Frontier, FTA, the L1-/S1-/TALE-style adapters, and Pooled-4; features are coarse cell-level summaries (provider, dataset, and training-only empirical accuracies). The held-out protocol was fixed before the reported evaluation and archived with the supplementary diagnostics. An earlier non-leakage-free analysis is excluded.

Table 4 gives the full result. Within this 15-cell design and feature set, the leakage-free selector does not improve over fixed Pooled-4.

## 8. Provider Heterogeneity and Transfer

Section 7 showed that FTA trails identical-pool plurality in aggregate. Here we ask whether that gap is explained by unreliable transfer of FTA’s failure-mined gates across providers. Table 5 reports per-cell rescue/regression counts for the two constituent gates (Low-Depth Guard / FIX-2; External-Consensus Fallback / FIX-4; Section 4.3).

### 8.1. Provider-stratified sign patterns

Mean FTA–Frontier deltas by provider: Cohere positive, Azure negative, Vertex and Fireworks mildly positive (Fireworks has three completed cells). Azure is the only provider with a negative mean; within Azure, harm concentrates on GSM8K/MATH-500 rather than StrategyQA or GPQA.



Table 5: Per-cell rescue/regression counts for FTA’s gates (FIX-2 / FIX-4). Applied switches: FIX-2 606, FIX-4 21, overlap 0. The table characterizes transfer of this failure-mined commit rule.

| Provider  | Data       | F2 trig. | F2 resc. | F2 regr. | F4 trig. | F4 resc. | F4 regr. |
|-----------|------------|----------|----------|----------|----------|----------|----------|
| Cohere    | GSM8K      | 63       | 36       | 9        | 3        | 3        | 0        |
| Cohere    | MATH-500   | 119      | 20       | 15       | 1        | 0        | 0        |
| Cohere    | GPQA-D     | 43       | 11       | 7        | 4        | 2        | 2        |
| Cohere    | StrategyQA | 0        | 0        | 0        | 2        | 0        | 2        |
| Azure     | GSM8K      | 26       | 2        | 18       | 0        | 0        | 0        |
| Azure     | MATH-500   | 60       | 7        | 31       | 2        | 0        | 0        |
| Azure     | GPQA-D     | 33       | 8        | 8        | 3        | 1        | 2        |
| Azure     | StrategyQA | 0        | 0        | 0        | 4        | 3        | 1        |
| Vertex    | GSM8K      | 17       | 4        | 11       | 0        | 0        | 0        |
| Vertex    | MATH-500   | 20       | 11       | 0        | 0        | 0        | 0        |
| Vertex    | GPQA-D     | 37       | 16       | 4        | 0        | 0        | 0        |
| Vertex    | StrategyQA | 0        | 0        | 0        | 0        | 0        | 0        |
| Fireworks | GSM8K      | 61       | 19       | 18       | 0        | 0        | 0        |
| Fireworks | MATH-500   | 125      | 26       | 9        | 0        | 0        | 0        |
| Fireworks | StrategyQA | 2        | 1        | 1        | 2        | 0        | 2        |

### 8.2. Which gate drives the movement?

Offline gate-only replay confirms FIX-2 accounts for essentially all FTA movement (606 applied switches vs. 21 for FIX-4; zero overlap; FIX-2-only accuracy matching FTA globally; Appendix G). Cell-level counts diverge: Azure×GSM8K FIX-2 yields 2 rescues against 18 regressions, and Azure×MATH-500 FIX-2 yields 7 rescues against 31 regressions, whereas the same Low-Depth Guard is net-positive on development-adjacent Cohere×GSM8K and on Fireworks×MATH-500 (Table 5).

Thus FTA is mechanistically FIX-2-dominated, and Azure mathematics cells are concrete transfer failures for this failure-mined commit rule under the evaluated provider shifts. FIX-2/FIX-4 were developed from Cohere×GSM8K failure diagnostics and frozen before the remaining cells, so the Azure and other non-development cells are the stronger transfer evidence (Section 11).

## 9. Compute Accounting and Instrumentation Semantics

Table 6 and Figure 4 evaluate the accounting semantics of equal Class A nominal budget  $B=6$ . Following Section 3, we separate (i) nominal logical

Table 6: Mean realized resources per evaluated Class A policy across 15 cells (nominal  $B=6$ ). Class B pool cost inherits up to four Class A pipelines (Table 2). USD uses a 2026-07-16 pricing snapshot. Frontier token/cost reconstruction status is documented in the table note.

| Class A method        | Tokens/ex. <sup>†</sup> | Latency (s) | Calls (nom.=6) | Cost/ex. <sup>†</sup> (\$) |
|-----------------------|-------------------------|-------------|----------------|----------------------------|
| Frontier              | 1287.0 <sup>†</sup>     | 16.87       | 5.38           | 0.00258 <sup>†</sup>       |
| L1-style adapter      | 728.7                   | 5.48        | 1.61           | 0.00133                    |
| S1-inspired adapter   | 1236.0                  | 8.80        | 2.85           | 0.00234                    |
| TALE-EP-style adapter | 727.3                   | 5.33        | 1.66           | 0.00129                    |

<sup>†</sup>Frontier tokens/cost are a documented lower bound (challenger-expansion-phase tokens are not recoverable from historical records without fabrication); calls, latency, and all adapter figures are complete. See Section 9.

actions, (ii) successful provider completions on the reported path, (iii) internal HTTP retries, (iv) generated-token fields, (v) observed latency, and (vi) estimated dollar cost. The columns are intentionally not collapsed: an instrumentation defect can change resource conclusions while leaving extracted answers and accuracy unchanged.

Table 6 reports unweighted means of per-cell means. Class B end-to-end cost inherits up to four Class A pipelines.

*Instrumentation audit and successful-call reconstruction.* A harness gap reset/read usage on Frontier’s outer generator only, omitting the challenger-expansion generator. On preserved validated Frontier records, successful calls reconstruct as outer successful calls plus `frontier_candidate_maturity` (maturity = challenger expansions + verifier calls). Mean successful calls rise from 2.78 to 5.38 of nominal  $B=6$  (example-weighted 5.44). Ratios versus the evaluated adapters: Frontier/L1-style  $2.13\times\text{--}5.18\times$ , Frontier/TALE-style  $2.03\times\text{--}5.51\times$ , Frontier/S1-inspired  $1.43\times\text{--}2.50\times$  (Frontier higher in every cell). Challenger-phase tokens were never persisted, so Frontier token and USD ratios are lower bounds ( $1.22\times\text{--}2.56\times$  vs. L1-style;  $1.13\times\text{--}2.93\times$  vs. TALE-style; mixed vs. S1-inspired); paid retries are likewise incomplete. Accuracy and McNemar/Holm tests are unchanged. The audit supports the claim that matched  $B=6$  did not produce matched realized call allocations (Appendix K).

Figure 4 is therefore a diagnostic for incomplete cost telemetry, not a definitive cost-efficiency frontier. The single-cell Azure $\times$ GSM8K SC comparison (Section 6.3) remains call/sample-budget-matched only.

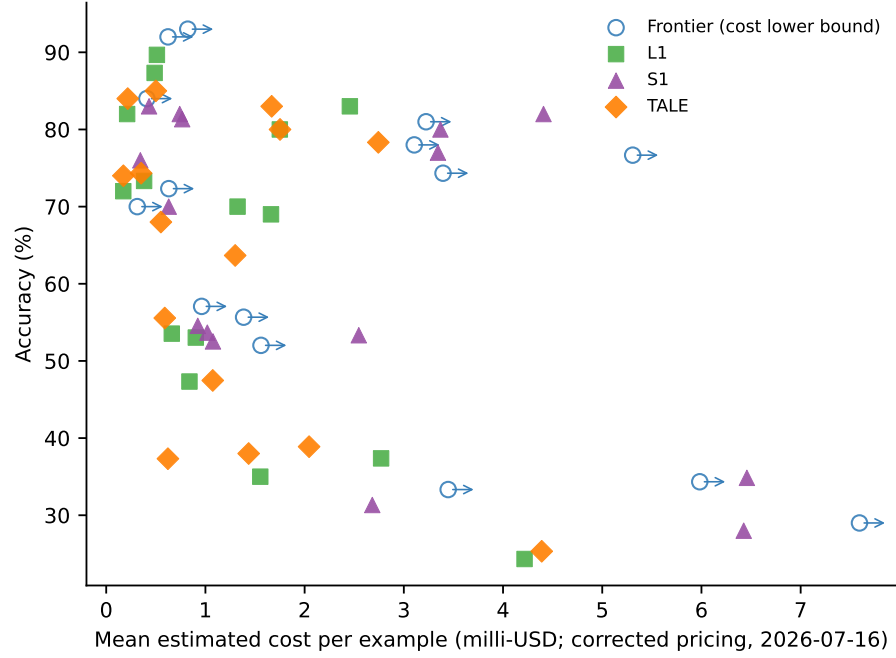


Figure 4: Incomplete-telemetry diagnostic for Class A accuracy and mean estimated cost per example (milli-USD). Open Frontier markers with rightward arrows indicate that challenger-phase token fields were not persisted, so Frontier cost positions are lower bounds rather than definitive Pareto or cost-efficiency rankings.

## 10. Offline Counterfactuals

Frozen-trace counterfactuals isolate attribution questions that aggregate accuracy cannot answer: whether challenger expansion changes outcomes relative to the incumbent, and whether repair scoring—orthogonal to allocation—moves commits or external rankings (Section 6). Gate-level transfer evidence remains in Section 8. Historical variants and diagnostics outside the current protocol are documented only in the supplementary material.

### 10.1. Incumbent versus full Frontier

Using stored incumbent answers from controller metadata, full Frontier beats the incumbent-only counterfactual in 9/15 cells and loses in 2/15 (Appendix H). This is a discovery-side non-inertness check under stored traces, not a matched-cost “no expansion” ablation.

### 10.2. Repair scoring counterfactual

Bounded repair is not credited to Frontier allocation. On the completed 15-cell matrix, Frontier’s scored answer differs from its repair-related field on 151/3394 rows; forcing the repair field for Frontier would yield 0 rescues and 68 regressions in the available counterfactual, while a repair-based external ranking counterfactual flips 0 external winners (Appendix B). These outcomes separate scoring convention from selection and accounting; primary tables retain the historical scoring convention.

## 11. Limitations

*Environments and budget.* Evidence covers four closed-API providers and four reasoning datasets at one nominal budget  $B=6$ , with 15/16 cells completed and Fireworks×GPQA protocol-blocked (Section 5). Aggregate identical-pool tests use  $n=3394$  paired examples, but transfer and cell-level policy claims rest on fewer independent environments; StrategyQA is underpowered for small deltas. Algorithm rankings at other budgets are not claimed. Natural Plan appears only as a supplementary pilot record.

*Baseline fidelity and development adjacency.* L1/S1/TALE rows are closed-API adapters, not official reproductions (Table 1). The main provenance qualifications are stated in Section 5; historical replay notes are supplementary. Metrics assume extractable, normalizable final answers.

*Resource telemetry.* Section 9 gives the resource-accounting detail. Token/USD fields remain lower bounds, paid retries are incomplete, and the study does not report FLOP matching or a full resource-matched matrix. Public-benchmark contamination in proprietary training data cannot be audited.

*Reproducibility and temporal validity.* Released artifacts support deterministic record replay (Section 5.4). They do not guarantee regeneration of identical proprietary outputs. Closed-API model identifiers do not necessarily provide immutable weight-level versioning, so provider-side updates may limit exact longitudinal regeneration. Historical component diagnostics are not ablations of the evaluated protocol.

## 12. Discussion

For settings like the one studied—multi-sample candidate pools, trace-conditioned selectors, provider transfer, or closed-API resource accounting—the protocol implies four minimum reporting practices for performance-measurement validity.

*Compare nontrivial selectors to identical-pool plurality.* Report accuracy against plurality on the same extracted answers and disclose pool-construction cost (Sections 7 and 7.3).

*Validate failure-mined rules across environments.* Label development-environment cells and report provider-stratified rescue/regression for the evaluated commit rule (Section 8).

*Report nominal  $B$  alongside componentwise  $\mathbf{r}$ .* Equal nominal budgets need not equalize calls, tokens, latency, or dollars; mark reconstruction status and lower-bound fields (Section 9).

*Separate discovery from selection, and keep blocked cells.* Jointly report DR and CSA, and retain protocol-blocked cells rather than imputing scores (Sections 6 and 5).

The common theme is semantic separation: nominal budgets, pool-construction cost, selector marginal cost, and measured token/USD fields answer different performance questions. Because this study is an evaluation of budgeted LLM reasoning on standard public benchmarks, its direct societal risks are those of LLM inference research generally; the most immediate

operational impact is that selective multi-generator protocols can increase or decrease aggregate compute depending on deployment choices.

### 13. Conclusion

Budgeted closed-API multi-call evaluation can change its defensible interpretation when discovery, selection, pool construction, and realized resources are conflated. This article contributes an accounting and attribution protocol for those settings. In comparable multi-sample, selector, transfer, and closed-service studies, efficiency and selection claims require the right resource layer and fixed candidate pools when selection is the object of evaluation.

### Acknowledgments

The author thanks his mother for her emotional support. The author is grateful to Professor Yiannis Koutis for his academic guidance and feedback. The author also thanks Anders Borum for providing complimentary lifetime access to Secure ShellFish, which supported remote access and development during this research.

### Funding

This research received in-kind computational support through API and cloud-service credits provided by the Cohere Labs Catalyst Grant Program, the Google Cloud Research Credits Program, Microsoft Azure for Students, and Fireworks AI. Cohere Labs provided USD 1,000 in API credits through its Catalyst Grant Program. These providers had no role in the study design; data collection, analysis, or interpretation; preparation of the manuscript; or the decision to submit the article for publication. The research received no other specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

### *Reproducibility Statement*

Released aggregate audit records, public manifests, source tables, and analysis scripts support aggregate-record verification of the reported tables and numerical invariants for all 15 completed cells; provenance tiers and replay limits are documented in Appendix I. The released artifacts do not

guarantee regeneration of identical proprietary API outputs, and live regeneration may incur API charges. The manifests define the public aggregate-record verification boundary for the reported analyses.

### **Declaration of competing interest**

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### **Data and code availability**

GSM8K, MATH-500, GPQA-Diamond, and StrategyQA are publicly available from their original sources. The public repository contains the evaluation code, frozen aggregate audit records, analysis scripts, manifests, and reproducibility instructions: <https://github.com/SoroushVahidi/frontier-allocation-for-budgeted-llm-inference>. The fixed public artifact commit is:

dfc9997d803199d699e23ee42cfe0777e6d78155.

The later public-release finalization commit is:

b5c8b1032205007ae5fd5fbc1fb4665e5aff0177.

### **Declaration of generative AI and AI-assisted technologies in the manuscript preparation process**

During manuscript preparation, the author used ChatGPT, Claude, and Cursor to assist with drafting and revising prose. Research-process uses of AI-assisted tools for code inspection, code modification, and analysis organization are described in Section 5.4. The author reviewed and edited the tool-assisted prose as needed and takes full responsibility for the content of the published article.

### **References**

- [1] M. Chen, J. Tworek, H. Jun, Q. Yuan, H. P. d. O. Pinto, et al., Evaluating large language models trained on code, arXiv preprint arXiv:2107.03374 (2021).

- [2] B. Brown, J. Juravsky, R. Ehrlich, R. Clark, Q. V. Le, C. Ré, A. Mirhoseini, Large language monkeys: Scaling inference compute with repeated sampling, arXiv preprint arXiv:2407.21787 (2024).
- [3] J. Hu, Oracle gap and signal fidelity: A fixed-pool diagnostic for test-time collaboration, arXiv preprint arXiv:2607.17531 (2026).
- [4] X. Wang, J. Wei, D. Schuurmans, Q. Le, E. Chi, S. Narang, A. Chowdhery, D. Zhou, Self-consistency improves chain of thought reasoning in language models, in: International Conference on Learning Representations (ICLR), 2023, arXiv:2203.11171.
- [5] C. Snell, J. Lee, K. Xu, A. Kumar, Scaling llm test-time compute optimally can be more effective than scaling model parameters, arXiv preprint arXiv:2408.03314, ICLR 2025 (2024).
- [6] P. Aggarwal, S. Welleck, L1: Controlling how long a reasoning model thinks with reinforcement learning, arXiv preprint arXiv:2503.04697 (2025).
- [7] N. Muennighoff, Z. Yang, W. Shi, X. L. Li, L. Fei-Fei, H. Hajishirzi, L. Zettlemoyer, P. Liang, E. Candès, T. Hashimoto, s1: Simple test-time scaling, in: Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics, 2025, pp. 20275–20321. doi:10.18653/v1/2025.emnlp-main.1025.  
URL <https://aclanthology.org/2025.emnlp-main.1025/>
- [8] T. Han, Z. Wang, C. Fang, S. Zhao, S. Ma, Z. Chen, Token-budget-aware llm reasoning, in: Findings of the Association for Computational Linguistics: ACL 2025, Association for Computational Linguistics, 2025, pp. 24842–24855. doi:10.18653/v1/2025.findings-acl.1274.  
URL <https://aclanthology.org/2025.findings-acl.1274/>
- [9] R. Jain, The Art of Computer Systems Performance Analysis: Techniques for Experimental Design, Measurement, Simulation, and Modeling, Wiley, 1991.
- [10] D. J. Lilja, Measuring Computer Performance: A Practitioner’s Guide, Cambridge University Press, 2000.



- [11] A. Georges, D. Buytaert, L. Eeckhout, Statistically rigorous java performance evaluation, in: Proceedings of the 22nd Annual ACM SIGPLAN Conference on Object-Oriented Programming Systems and Applications, ACM, 2007, pp. 57–76. doi:10.1145/1297027.1297033.
- [12] M. A. Alomrani, Y. Zhang, D. Li, Q. Sun, S. Pal, Z. Zhang, Y. Hu, R. D. Ajwani, A. Valkanas, R. Karimi, P. Cheng, Y. Wang, P. Liao, H. Huang, B. Wang, J. Hao, M. Coates, Reasoning on a budget: A survey of adaptive and controllable test-time compute in llms, arXiv preprint arXiv:2507.02076 (2025).
- [13] A. V. Papadopoulos, L. Versluis, A. Bauer, N. Herbst, J. von Kistowski, A. Ali-Eldin, C. L. Abad, J. N. Amaral, P. Tuma, A. Iosup, Methodological principles for reproducible performance evaluation in cloud computing, IEEE Transactions on Software Engineering 47 (8) (2021) 1528–1543. doi:10.1109/TSE.2019.2927908.
- [14] H. Guo, H. Cao, J. He, X. Liu, Y. Shi, POBO: Safe and optimal resource management for cloud microservices, Performance Evaluation 162 (2023) 102376. doi:10.1016/j.peva.2023.102376.
- [15] A. Ali, X. Ma, S. Zawad, P. Aditya, I. E. Akkus, R. Chen, L. Yang, F. Yan, Enabling scalable and adaptive machine learning training via serverless computing on public cloud, Performance Evaluation 167 (2025) 102451. doi:10.1016/j.peva.2024.102451.
- [16] P. Liang, R. Bommasani, T. Lee, D. Tsipras, D. Soylu, M. Yasunaga, Y. Zhang, D. Narayanan, Y. Wu, A. Kumar, et al., Holistic evaluation of language models, Transactions on Machine Learning Research (2023). doi:10.48550/arXiv.2211.09110.  
URL <https://arxiv.org/abs/2211.09110>
- [17] C. White, S. Dooley, M. Roberts, A. Pal, B. Feuer, S. Jain, R. Shwartz-Ziv, N. Jain, K. Saifullah, S. Dey, Shubh-Agrawal, S. S. Sandha, S. Naidu, C. Hegde, Y. LeCun, T. Goldstein, W. Neiswanger, M. Goldblum, Livebench: A challenging, contamination-limited LLM benchmark, in: International Conference on Learning Representations (ICLR), 2025, spotlight; arXiv:2406.19314. doi:10.48550/arXiv.2406.19314.  
URL <https://arxiv.org/abs/2406.19314>

- [18] I. Mirzadeh, K. Alizadeh, H. Shahrokhi, O. Tuzel, S. Bengio, M. Farajtabar, GSM-Symbolic: Understanding the limitations of mathematical reasoning in large language models, in: International Conference on Learning Representations (ICLR), 2025. doi:10.48550/arXiv.2410.05229.  
URL <https://arxiv.org/abs/2410.05229>
- [19] W. Kwon, Z. Li, S. Zhuang, Y. Sheng, L. Zheng, C. H. Yu, J. E. Gonzalez, H. Zhang, I. Stoica, Efficient memory management for large language model serving with PagedAttention, in: Proceedings of the 29th ACM Symposium on Operating Systems Principles, Association for Computing Machinery, 2023, pp. 611–626. doi:10.1145/3600006.3613165.  
URL <https://dl.acm.org/doi/10.1145/3600006.3613165>
- [20] Z. Zhai, B. Li, B. Xiao, M. Li, X. Wang, Adaptive test-time compute allocation for reasoning llms via constrained policy optimization, arXiv preprint arXiv:2604.14853 (2026).
- [21] S. Huang, H. Wang, W. Zhong, Z. Su, J. Feng, B. Cao, Y. R. Fung, AdaCtrl: Towards adaptive and controllable reasoning via difficulty-aware budgeting, Transactions on Machine Learning Research, arXiv:2505.18822 (2026).
- [22] D. Patel, Anytime verified agents: Adaptive compute allocation for reliable LLM reasoning under budget constraints, Transactions on Machine Learning Research (2026).
- [23] X. Chen, R. Aksitov, U. Alon, J. Ren, K. Xiao, P. Yin, S. Prakash, C. Sutton, X. Wang, D. Zhou, Universal self-consistency for large language model generation, arXiv preprint arXiv:2311.17311 (2023).
- [24] P. Aggarwal, A. Madaan, Y. Yang, Mausam, Let’s sample step by step: Adaptive-consistency for efficient reasoning and coding with LLMs, in: Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics, 2023, pp. 12375–12396. doi:10.18653/v1/2023.emnlp-main.761.
- [25] Y. Li, P. Yuan, S. Feng, B. Pan, X. Wang, B. Sun, H. Wang, K. Li, Escape sky-high cost: Early-stopping self-consistency for multi-step

- reasoning, in: International Conference on Learning Representations (ICLR), 2024, arXiv:2401.10480.
- [26] X. Wang, S. Feng, Y. Li, P. Yuan, Y. Zhang, C. Tan, B. Pan, Y. Hu, K. Li, Make every penny count: Difficulty-adaptive self-consistency for cost-efficient reasoning, in: Findings of the Association for Computational Linguistics: NAACL 2025, Association for Computational Linguistics, 2025, pp. 6919–6932. doi:10.18653/v1/2025.findings-naacl.383.
  - [27] J. Kim, N. Yang, K. Min, K. Jung, Reliability-aware adaptive self-consistency for efficient sampling in LLM reasoning, in: Findings of the Association for Computational Linguistics: ACL 2026, Association for Computational Linguistics, 2026, pp. 21575–21590. doi:10.18653/v1/2026.findings-acl.1085.  
URL <https://aclanthology.org/2026.findings-acl.1085/>
  - [28] Z. Kang, X. Zhao, D. Song, Scalable best-of-N selection for large language models via self-certainty, in: Advances in Neural Information Processing Systems, 2025, arXiv:2502.18581.
  - [29] H. Lightman, V. Kosaraju, Y. Burda, H. Edwards, B. Baker, T. Lee, J. Leike, J. Schulman, I. Sutskever, K. Cobbe, Let’s verify step by step, arXiv preprint arXiv:2305.20050 (2023).
  - [30] L. Chen, M. Zaharia, J. Zou, FrugalGPT: How to use large language models while reducing cost and improving performance, arXiv preprint arXiv:2305.05176 (2023).
  - [31] D. Ding, A. Mallick, C. Wang, R. Sim, S. Mukherjee, V. Ruhle, L. V. S. Lakshmanan, A. H. Awadallah, Hybrid LLM: Cost-efficient and quality-aware query routing, in: International Conference on Learning Representations (ICLR), 2024.
  - [32] I. Ong, A. Almahairi, V. Wu, W.-L. Chiang, T. Wu, J. E. Gonzalez, M. W. Kadous, I. Stoica, RouteLLM: Learning to route LLMs with preference data, in: International Conference on Learning Representations (ICLR), 2025, arXiv:2406.18665.
  - [33] S. Yao, D. Yu, J. Zhao, I. Shafran, T. L. Griffiths, Y. Cao, K. Narasimhan, Tree of thoughts: Deliberate problem solving with large

- language models, *Advances in Neural Information Processing Systems*, arXiv:2305.10601 (2023).
- [34] S. Russell, E. Wefald, *Do the Right Thing: Studies in Limited Rationality*, MIT Press, 1991.
  - [35] K. Cobbe, V. Kosaraju, M. Bavarian, M. Chen, H. Jun, L. Kaiser, M. Plappert, J. Tworek, J. Hilton, R. Nakano, C. Hesse, J. Schulman, Training verifiers to solve math word problems, *arXiv preprint arXiv:2110.14168* (2021).
  - [36] D. Hendrycks, C. Burns, S. Kadavath, A. Arora, S. Basart, E. Tang, D. Song, J. Steinhardt, Measuring mathematical problem solving with the math dataset, *arXiv preprint arXiv:2103.03874* (2021).
  - [37] D. Rein, B. L. Hou, A. C. Stickland, J. Petty, R. Y. Pang, J. Dirani, J. Michael, S. R. Bowman, Gpqa: A graduate-level google-proof q&a benchmark, *arXiv preprint arXiv:2311.12022* (2023).
  - [38] M. Geva, D. Khashabi, E. Segal, T. Khot, D. Roth, J. Berant, Did aristotle use a laptop? a question answering benchmark with implicit reasoning strategies, *Transactions of the Association for Computational Linguistics (TACL)* (2021).

This appendix supplies disclosure and provenance detail summarized in the main text: gold-free features, repair and baseline fidelity, trust tiers, statistical testing, pseudocode, offline counterfactuals, and compute detail. Historical negative variants, pilot surfaces, and partial same-model checks are moved to the supplementary material because they are outside the evaluated protocol.

## Appendix A. Gold-Free Runtime Features

Every quantity used by Frontier (incumbent support, challenger support, override margin, and maturity) and by FTA (trace trigger, external agreement) is computed from generated candidates only. Gold labels are offline-only for accuracy and discovery/selection diagnostics. Runtime leakage checks found no gold-dependent runtime features. Researcher-adaptation concerns remain separate from runtime leakage (Section 11).

## Appendix B. Bounded Answer Repair

Bounded repair re-ranks completed branch nodes by canonicalized answer support. Surfacing overrides repair with a controller-committed answer when present. Frontier almost always supplies a committed answer, so repair is computed but discarded for scoring; L1/S1/TALE typically score the repair-layer output. Offline quantification on the 15-cell matrix: Frontier’s scored answer differs from the repair-related field on **151** of 3,394 Frontier rows. Among Frontier differs, hypothetical repair application yields 0 rescues and 68 regressions in this reconstruction (remainder wrong $\leftrightarrow$ wrong / neutral). External `repair_answer_raw` vs. scored often differs textually (462 rows) largely due to canonicalization; forcing a repair-based external ranking counterfactual produced **0** external winner flips. Asymmetry remains real; external cell winners appear ranking-stable under this counterfactual check.

## Appendix C. Baseline Fidelity (Full)

**L1-style adapter.** Original contribution is LCPO training; we use only the length-instruction prompt surface (closer to the paper’s untrained budget-prompt ablation).

**S1-inspired adapter.** Original suppresses end-of-thinking in a local token stream; we approximate via repeated chat “Wait” continuations.

**TALE-EP-style adapter.** We target TALE-EP prompt injection; replace zero-shot budget estimation with a deterministic character-length heuristic to preserve the  $B=6$  call-budget protocol. TALE-PT is not reproduced.

## Appendix D. Provenance and Trust Tiers

Tier A: Cohere/Azure/Vertex $\times$ GSM8K (strongest draw and scoring provenance under the released manifests). Tier B: remaining completed cells. Three cells had historical duplicate-row noise; deduplication changed 0 accuracies. Azure $\times$ GPQA FTA: offline selector replay on stored generations. Fireworks $\times$ GPQA: protocol-blocked (BLOCKED\_PROTOCOL\_NONCONVERGENCE). For some Tier B cells the frozen evaluated subset is preserved even when the complete historical draw procedure cannot be reconstructed (Section 5.4).

## Appendix E. Statistical Testing

Paired exact McNemar tests; Holm–Bonferroni within relevant comparison families. Class A and Class B are not Holm-mixed as if they shared a budget unit. Bootstrap vs. McNemar disagreement on Cohere $\times$ GSM8K FTA-vs-L1 (high ties) is reported historically; we do not revive the weaker significance claim. For Pooled-4 vs. FTA, the stratified within-cell paired bootstrap on the frozen  $n=3394$  rows gives a 95% percentile interval  $[+0.74, +2.33]$  pp. Provider- and dataset-level leave-one-out deletions are sensitivity checks on the same sample, not independent replications.

## Appendix F. Procedures (Pseudocode)

### Appendix F.1. Frontier (Class A reserved-attempt full-expansion policy)

Table F.7 lists the fixed public parameters for the reported matrix cells.

*Frontier procedure.*

1. **Input:** instance  $x$  and nominal budget  $B=6$ .
2. Generate two reserved direct-chain attempts with distinct prompt styles.
3. Group extracted answers by the public gold-free canonicalization rule and choose the incumbent by plurality; ties break by first-observed answer group.

Table F.7: Frontier public parameters for all reported matrix cells. These are fixed across providers and datasets.

| Quantity                          | Executed value                                                          |
|-----------------------------------|-------------------------------------------------------------------------|
| Nominal budget                    | $B=6$                                                                   |
| Reserved direct attempts          | 2 prompt styles                                                         |
| Answer grouping                   | Gold-free canonical answer key; first-observed tie-break                |
| Template early gate               | support $\geq 2.0$ , margin $\geq 2.0$ , entropy $\leq -1.0$ (sentinel) |
| Gate status in reported runs      | Disabled by sentinel; never fired                                       |
| Override maturity threshold       | challenger maturity $\geq 2$                                            |
| Override support-margin threshold | challenger support – incumbent support $\geq 1$                         |
| Override guard                    | no override when incumbent appears in challenger-supported groups       |

4. Allocate the remaining nominal budget, if any, to challenger generation and verification. The template retained an early-commit gate, but the evaluated thresholds are sentinel values (support threshold 2.0 with vote-share support in  $[0, 1]$ ), so early commitment never fires and is not part of the effective evaluated policy.
5. Override the incumbent with the challenger iff a challenger exists, differs from the incumbent, has support greater than one, has maturity  $\geq 2$ , has support margin  $\geq 1$ , and the incumbent is not already represented among challenger-supported groups.
6. **Output:** the committed answer. All thresholds and tie rules are fixed across cells; features are gold-free.

*Action semantics.* For Frontier, one nominal action is one direct-chain attempt or one challenger expansion step, each corresponding to one successful provider response on the reported path. For L1 and TALE, one action is one length/budget-conditioned generation attempt; for S1, it is the initial generation or a forced “Wait” continuation. Canonicalization, plurality, repair ranking, FTA gates, and Pooled-4 voting issue no model calls. Internal HTTP retries are counted in retry telemetry but not as new logical actions unless a new successful response is produced.

*Pooled-4 (Class B consensus).*

1. **Input:** committed answers of Frontier, L1, S1, and TALE for instance  $x$ .

2. Let  $a^*$  be a plurality mode (agreement  $\geq 2$ ).
3. If no unique mode, return Frontier’s answer (deterministic tie-break).
4. **Output:**  $a^*$ . No additional model calls.

*FTA (Class B gated deferral).*

1. **Input:** Frontier answer and trace metadata; L1/S1/TALE answers.
2. If Low-Depth Guard (FIX-2) triggers on a single-weak-branch (or support-zero) signature, return majority among  $\{L1, S1, TALE\}$  (agreement  $\geq 2$ ; priority  $TALE \succ S1 \succ L1$ ).
3. Else if External-Consensus Fallback (FIX-4) triggers (direct incumbent agreement and unanimous external disagreement with Frontier), return the unanimous external answer.
4. Else return Frontier’s answer.
5. Gates are mutually exclusive by construction; at most one fires.

## Appendix G. Gate Details and Ablation

FIX-2 and FIX-4 are non-overlapping by construction. Applied switches: 606 / 21; overlap 0. FIX-2-only accuracy equals FTA globally; FIX-4-only  $\approx$  Frontier. Azure harm is FIX-2-driven (Section 8).

## Appendix H. Incumbent Ablation

Using stored incumbent-only answers from controller metadata: Frontier beats incumbent in 9/15 cells; incumbent better in 2/15. Large gains on several MATH cells. Caveat: metadata completeness varies. This is an offline comparison against stopping after the reserved attempts, not a live early-stopping policy and not a matched-cost alternative.

## Appendix I. Reproducibility Records

A public claim–evidence map links numerical claims to tracked aggregate audit records, source tables, analysis scripts, and package checksums for the Azure true-SC comparison, held-out selector, and 15-cell matrix summaries. The public artifacts support aggregate-record verification of the reported numerical invariants, not identical proprietary regeneration or every historical subset draw (Section 5.4). Live reruns would require provider credentials and paid API calls and are not part of this manuscript.



## Appendix J. Supplementary Diagnostics

The descriptive discovery–selection scatter diagnostic is provided in the supplementary material; the tested transfer evidence used in the main paper is reported in Section 8.

## Appendix K. Additional Compute Detail

Frontier/L1: tokens  $1.10\times-2.57\times$  (lower bound), calls  $2.13\times-5.18\times$  (exact), latency  $2.11\times-4.39\times$ . Frontier/TALE: tokens  $1.01\times-2.65\times$  (lower bound), calls  $2.03\times-5.51\times$  (exact), latency  $1.80\times-6.77\times$ . Frontier/S1: tokens mixed (lower bound); calls  $1.43\times-2.50\times$  (exact; Frontier higher every cell); latency higher every cell. Dollar figures use recomputed pricing from measured tokens (2026-07-16 snapshot). Tokens, USD from incomplete token fields, and paid retries remain lower bounds.

*Pooled-4 tie-rule sensitivity.* On 3,259 complete four-method quads, 88.0% have a unique plurality and 12.0% fall back. Accuracy under Frontier-fallback (67.8% on this reconstruction) moves to 66.7%/66.5% under two predeclared alternatives ( $\leq 1.3$  pp); none selected post hoc. This reconstruction excludes 135 incomplete-quad rows and is for tie-sensitivity only, not a replacement of the 66.53%/n=3394 aggregate.